

Yutong (Richie) Jin

Auckland, New Zealand | +64 027 292 2743 | richiejyt@gmail.com
github.com/R1chi33333 | r1chi33333.github.io | linkedin.com/in/yutong-jin-a1743b291

SUMMARY

Master of Engineering graduate who builds and ships full-stack and machine-learning products end to end — production ML for healthcare and energy, tested web apps running live on cloud infrastructure, and multi-modal AI agents (LLM, vision, speech). Fluent across C#/.NET, Python and TypeScript/React, with a research background in human-agent interaction. Auckland-based, open to work.

TECHNICAL SKILLS

Languages: C#, Python, TypeScript / JavaScript, Java, SQL, R, Swift

Frontend: React, Next.js, Tailwind CSS, HTML / CSS

Backend & Data: ASP.NET Core, FastAPI, Node.js, PostgreSQL, Drizzle, SQLAlchemy, REST & WebSocket

ML & AI: PyTorch, XGBoost, scikit-learn, LLM / RAG, pgvector, YOLOv8, Ollama, transformers.js

DevOps & Tools: Docker, Kubernetes, Git, GitHub Actions (CI/CD), Linux, Unity, Vercel / Railway / Render

Spoken: English (Fluent), Korean (Fluent), Chinese (Native)

PROFESSIONAL EXPERIENCE

HITLab NZ, University of Canterbury

Research Assistant

Christchurch, New Zealand

Nov 2025 – Jan 2026

- Integrated multi-modal AI models (LLM, VLM, TTS, STT) into a Unity-based educational project for interactive learning.
- Built a full-stack architecture with a Python backend and responsive Unity frontend; documented the codebase to onboard beginners to C#.

WEL Networks

Software Engineer Intern

Christchurch, New Zealand

Jun 2025 – Jul 2025

- Built a Python inverter-compliance system over 6M+ smart-meter records, implementing Export Limit, Volt-Watt and Volt-Var detection.
- Delivered an automated tool that replaced manual verification, enabling real-time monitoring of grid stability.

Shenzhen Institute of Advanced Technology (SIAT), CAS

Machine Learning Engineer

Shenzhen, China

Sep 2022 – Oct 2023

- Engineered a core-body-temperature prediction system at 95% accuracy, deployed for real-time monitoring.
- Built an ECG/IMU exercise-fatigue detection platform (85.46%); optimized models (XGBoost, SVR, Neural Networks) for edge, cutting inference time by 40%.

SOFTWARE PROJECTS

nz-netpay · C# / ASP.NET Core 8, xUnit, Docker

[Live](#) · [Code](#)

REST API for NZ take-home pay (PAYE, ACC, KiwiSaver, student loan) from official IRD/ACC rates; 38 tests, 97% engine coverage, CI/CD, deployed live.

TenancyMate · Next.js, RAG, pgvector

[Live](#) · [Code](#)

Retrieval-augmented Q&A over NZ tenancy law with section-level citations; published a 47-question eval with zero hallucinated citations.

NZ Housing Observatory · React, MapLibre, GitHub Actions

[Live](#) · [Code](#)

Interactive dashboard of 46 years of NZ house prices and rents over a tested pipeline that refreshes monthly via scheduled CI.

FlatSplit · Next.js, PostgreSQL, NextAuth

[Live](#) · [Code](#)

Full-stack expense-splitting app for NZ flats with a minimum-transfer settlement algorithm and bank-CSV import.

RESEARCH & ACADEMIC PROJECTS

Master's Thesis — Intelligent VR Games with Multi-Modal AI Agents

2025 – 2026

- Built three VR games each with an autonomous LLM+STT+TTS agent; ran a between-subjects study (26 participants) on agent identity continuity, analysed with ANCOVA in Python, finding significant gains in rapport and emotional connectedness.

COVID-19 Vaccine Response Analysis — with Dr. Yunpeng Cai, SIAT

Shenzhen, China

- Analysed 50,000+ vaccination records with ensemble ML (Random Forest, Neural Networks) to surface vaccine-symptom correlations.

EDUCATION

University of Canterbury — M.E. Human Interface Technology (GPA 7.63)

Feb 2025 – Apr 2026

Yanbian University — B.S. Computer Science

Jun 2017 – Sep 2021